

v22 – Common source-native boundary refinement, source C_{11} module, primitive-index 121 recurrence witnesses, and exceptional incidence on $\widehat{X}$

Formal HT+NT continuation

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Abstract

v22 retains v21 **CLOSED/APPEND-ONLY**. The three regular polygonal boundary carriers of lengths 78, 66, 62 admit a connected common source-native refinement obtained by gluing exact identical historical-edge subpaths along the maximum spanning tree of their overlap graph. The common graph has $(V, E, b_1) = (195, 197, 3)$, so no extra common one-cycle is introduced, and all three occurrencewise collapses factor through it. The induced relative complexes in degrees $3 \rightarrow 2$ are both the identity presentation $\mathbb{Z}^3 \xrightarrow{I_3} \mathbb{Z}^3$, hence relative $H_3 = H_2 = 0$ before and after collapse. The interaction C_{11} acquires its first source-native nontrivial action: the degree-five factor of $x^{44} - x^9 - 1$ has root order 121, so its eleventh power gives a faithful C_{11} action on a five-dimensional $\mathbb{F}_3$ module. An exhaustive factor census for all $2 \leq k \leq 60$ finds 59 exact recurrence occurrences of primitive-index word $(1, 2, 1)$; the first is $(k, v) = (9, 2)$. Finally the five exceptional source components do not map one-to-one to the five singular components of $\widehat{X}$: E_T, E_U, E_0 dominate S_1, S_2, S_5 , while E_a, E_b dominate the two Λ wall substrata inside S_1, S_2 ; S_3, S_4 occur only as lower-dimensional specializations. Crossing a generic real Λ wall has identity sheet monodromy; the nontrivial C_2 monodromy belongs to the branch divisor $H = 0$, not to $\Lambda = 0$ itself.

1 Frozen parent and documentary typing

v21 remains immutable with token

V21_STEINER_CW_SPLIT_V4_FACTOR121_XHAT_CHAMBER_PASS.

The retained polygonal cycles are

$$C_{78}, \quad C_{66}, \quad C_{62},$$

with occurrencewise source-edge collapses $\pi_{25}, \pi_{40}, \pi_{57}$. The HT+NT guard remains unchanged: numerical equality, common labels, or equal ternary words do not merge carriers without an explicit structure-preserving map. The greedy/lazy lineage is used in the same sense: a bridge is promoted only after a state/action morphism is materialized.

2 Common source-native refinement of the three boundary carriers

For each pair of v21 cycles, search cyclically for the longest contiguous sequence of *identical oriented historical source-edge tokens*. The exact overlap graph is

pair	length	maximal source-identical path
$25 \leftrightarrow 40$	5	$(+200, +280, +364, -281, -201)$
$25 \leftrightarrow 57$	4	$(-200, +204, +288, -206)$
$40 \leftrightarrow 57$	1	$(+200)$.

Choose the maximum spanning tree of this three-vertex weighted graph, hence the first two overlaps. Glue precisely those occurrence paths. Every gluing identifies only vertices with the same source object and only edges with the same oriented historical label. No two distinct vertices or edges of any one polygon are identified, so the three cycles remain embedded.

The quotient graph $\widetilde{\Gamma}$ is connected and satisfies

$$|V(\widetilde{\Gamma})| = 195, \quad |E(\widetilde{\Gamma})| = 197, \quad b_1(\widetilde{\Gamma}) = 3.$$

Thus its cycle rank is exactly the sum of the three carrier-cycle ranks: the common refinement introduces no additional one-cycle. Each collapse factors as

$$C_{N_v} \hookrightarrow \tilde{\Gamma} \xrightarrow{\rho} G_{\text{cover}}^{(1)}, \quad \pi_v = \rho \circ \iota_v,$$

where every refined edge maps homeomorphically to its exact historical source edge. The remaining $40 \leftrightarrow 57$ edge overlap is intentionally not glued: after the spanning-tree gluing it would raise b_1 from 3 to 4 and create an artificial common cycle.

Hence

$$\boxed{\tilde{\Gamma} = \text{PROVEN-COMMON-SOURCE-NATIVE-REFINEMENT}}$$

in the declared path-glued occurrencewise class. Maximality refers only to overlap along a spanning tree of exact identical source paths; arbitrary graph refinements remain outside this claim.

3 Relative H_2/H_3 before any primal promotion

Take the retained cover/base complex together with the common boundary carrier as the relative base. The only new relative cells are

$$\tilde{c}_{25}, \tilde{c}_{40}, \tilde{c}_{57} \quad \text{and} \quad \tilde{\sigma}_{25}, \tilde{\sigma}_{40}, \tilde{\sigma}_{57}.$$

Modulo the base, each Bianchi 3-cell has boundary exactly its corresponding connector 2-cell. Therefore the relative cellular complex is

$$0 \longrightarrow \mathbb{Z}^3 \xrightarrow{I_3} \mathbb{Z}^3 \longrightarrow 0.$$

The v19/v21 collapsed relative thickening has the same degree-3 $\rightarrow$ 2 presentation, and the collapse chain map is I_3 in both degrees. Consequently

$$\boxed{H_3^{\text{rel}} = 0, \quad H_2^{\text{rel}} = 0}$$

on source and target, integrally and after every coefficient reduction. The induced maps are the unique $0 \rightarrow 0$ isomorphisms. This is a genuine **PASS**, but it is a negative promotion result:

$$\boxed{\text{no new relative } H_2/H_3 \text{ class supports a primal promotion.}}$$

4 A source-native C_{11} module from the recurrence factor carrier

Retain

$$\text{coker } \hat{d}_{\text{int}} \simeq C_{11} \oplus \hat{V}_4,$$

with the interaction generator represented by $\gamma = [(2, 2, 4)]$ of exact order 11. For

$$f_{44,9}(x) = x^{44} - x^9 - 1 \in \mathbb{F}_3[x],$$

the unique degree-five irreducible factor is

$$\boxed{g_9(x) = x^5 - x^4 - x^2 - 1.}$$

In

$$M_9 = \mathbb{F}_3[x]/(g_9), \quad \alpha = [x],$$

one has $\text{ord}(\alpha) = 121$. Hence

$$\beta = \alpha^{11}, \quad \text{ord}(\beta) = 11,$$

and multiplication by β is a faithful source-native C_{11} action on the five-dimensional $\mathbb{F}_3$ module M_9 . In the basis $(1, x, x^2, x^3, x^4)$ its generator matrix is

$$B_9 = \begin{pmatrix} 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 1 & 2 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 & 2 \end{pmatrix} \pmod{3}, \quad B_9^{11} = I,$$

with no smaller positive power equal to I .

The reflected tap $v = 35$ has

$$g_{35}(x) = x^5 + x^3 + x + 1 = -x^5 g_9(-1/x),$$

and if $\alpha' = -\alpha^{-1}$ then $\text{ord}(\alpha') = 242$ and

$$\beta' = (\alpha')^{22} = \beta^{-2} = \beta^9.$$

Thus negative reciprocal transport acts on the recovered C_{11} by the explicit automorphism $r \mapsto 9r$ modulo 11. Sending γ to multiplication by β gives

$$C_{11}^{\text{int}} \curvearrowright M_9 = \text{PROVEN-TYPED-SOURCE-NATIVE-MODULE-ADAPTER.}$$

The Stone no-go is retained: this does not create a nontrivial C_{11} action on $\langle \mu_{11} \rangle$ or on $\text{Aut}(\mathbb{Z}/24)$.

5 Extended primitive-index census: 121 is now source-realized

The v21 status was ZERO in its retained factor corpus but globally NT. v22 performs an exhaustive scan of every

$$f_{k,v}(x) = x^k - x^v - 1 \in \mathbb{F}_3[x], \quad 2 \leq k \leq 60, \quad 1 \leq v < k.$$

All 1770 polynomials are factored exactly. For every case having exactly three irreducible factors, the root orders and primitive indices

$$\delta_d = \frac{3^d - 1}{\text{ord}(\alpha_d)}$$

are computed exactly. There are 481 three-factor cases and

$$59 \text{ occurrences of } (1, 2, 1).$$

The lexicographically first witness is

$$(k, v) = (9, 2)$$

with factor data

d	$g_d(x)$	$\text{ord}(\alpha_d)$	δ_d
1	$x + 1$	2	1
3	$x^3 + x^2 + x - 1$	13	2
5	$x^5 + x^4 - x^3 + 1$	242	1.

Therefore

$$(1, 2, 1) = 121 \text{ is now PROVEN-EXISTS in the recurrence carrier.}$$

This closes the previous global NT existence question. It does *not* identify the mechanism with the roof identity $11_3^2 = 121_3$, and it does not produce an $AGL(1, 3)$ action on a single recurrence carrier. The four palindrome words containing zero remain type-incompatible with positive primitive-index coordinates, so full AGL completion remains NOT-FORMED.

6 Exceptional incidence on the normal quadratic cover

Retain

$$\widehat{X} : \quad t^2 = H, \quad H = 2025a^2b^2 + 6abc(q - 266p) + c^2(2p - q)^2,$$

and

$$\Lambda = (37p - 2q)(118p + 7q).$$

The five singular components remain

$$\begin{aligned} S_1 : t = a = c = 0, & & S_2 : t = b = c = 0, \\ S_3 : t = a = p = q = 0, & & S_4 : t = b = p = q = 0, \\ S_5 : t = a = b = q - 2p = 0. \end{aligned}$$

Exact substitution of the five v21 exceptional source components gives the dominance graph

$$\begin{array}{l|l} E_T : T = \frac{1}{2} & S_1 \text{ (dominant)} \\ E_U : U = \frac{1}{2} & S_2 \text{ (dominant)} \\ E_a : x + 2z = 5y + 3z = 0 & W_a := S_1 \cap \{37p - 2q = 0\} \\ E_b : x + 4y = 5y + 3z = 0 & W_b := S_2 \cap \{118p + 7q = 0\} \\ E_0 : x + 2z = x + 4y = 0 & S_5 \text{ (dominant).} \end{array}$$

In particular, E_a and E_b land exactly over the two factors of the Λ discriminant wall. No irreducible exceptional component dominates S_3 or S_4 ; those are lower-dimensional specialization/intersection strata of the divisorial contractions. Therefore a putative five-by-five exceptional/singular bijection is

$$\text{REFUTED-TYPED.}$$

Real sheet monodromy across $\Lambda = 0$

The walls $37p - 2q = 0$ and $118p + 7q = 0$ govern the signature of the binary quadratic form $H(u, c)$, $u = ab$, but they are not themselves the branch divisor. On every path contained in $H > 0$, the sign of t is continuous, so the two labels

$$t = +\sqrt{H}, \quad t = -\sqrt{H}$$

cannot exchange without meeting $t = 0$. A transverse certificate is obtained by setting $u = 1, c = 0$, for which

$$H = 2025, \quad t = \pm 45$$

independently of (p, q) ; both Λ walls can therefore be crossed through the smooth real locus with no exchange. Hence

generic real Λ -wall monodromy = id.

If a path enters a sector with $H < 0$, the real lift simply ceases to exist; this is not a transposition. The nontrivial C_2 monodromy belongs instead to a loop around the genuine branch divisor $H = 0$ in the complexified cover. Real wall transport and complex branch monodromy remain **SEPARATED**.

7 Gate ledger

claim	status
v21 state and three minimal polygons retained	PASS
common source-native $\tilde{\Gamma}$, $(V, E, b_1) = (195, 197, 3)$	PASS
maximal overlap in declared tree-glued path class	PASS
relative H_2/H_3 before/after collapse	0/0; induced maps PASS
primal promotion from these relative groups	NOT-OPENED
source-native recurrence C_{11} module	PASS
C_{11} action on Stone μ_{11} carrier	REFUTED-TYPED retained
primitive-index recurrence word 121	PASS (59 witnesses for $k \leq 60$)
full recurrence $AGL(1, 3)$ palindrome carrier	NOT-FORMED
exceptional/singular incidence on $\hat{X}$	PASS
five-by-five exceptional/singular bijection	REFUTED-TYPED
real Λ -wall monodromy	PASS: identity
complex $H = 0$ branch monodromy	SEPARATED C_2
C_2^5	NOT-OPENED
T_7	SEPARATED scalar fibre

The atomic publication state after destructive replay is

CLOSED / APPEND-ONLY

with tokens

V22_COMMON_REFINEMENT_C11_MODULE_FACTOR121_XHAT_INCIDENCE_PASS

and

V22_POSTSEAL_REPLAY_PASS.

8 Next safe frontier

The next gate should preserve the common refinement and now test its *mapping-cylinder* geometry rather than reopening the already-zero relative top homology. A safe v23 target is:

v23 – retain $\tilde{\Gamma}$ and construct the explicit cellular mapping cylinder $M(\rho)$;
 compute absolute H_1, H_2, H_3 of the regularized pushout and compare to the v19 thickening;
 test whether the three connector disks admit source-compatible 2-cell syzygies before any primal merge;
 C_{11} : classify the 59 recurrence 121 witnesses by the induced C_{11} factor-module data;
 test whether the 212 and 121 witnesses are related by a declared recurrence operation;
 $\hat{X}$: compute the full intersection poset of S_i, W_a, W_b and local normal forms along S_3, S_4 ;
 separate real wall transport from complex branch monodromy throughout;
 C_2^5 remains NOT-OPENED, and T_7 remains SEPARATED.